WORKED SOLUTION

STEP III 2002, Question 4

Pure • Diophantine Inequalities

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/02-s3-q4

The key idea

The equation $x^3 + x^2 = y^3 - y^2$ pits a quantity that grows fast against one that is held back by the $-y^2$, so y has to "overtake" x, but only by less than 1. Those two inequalities squeeze y strictly between x and $x+1$, and no integer fits in that gap. For the last part, the extra solutions can only come from zero or negative values, and a neat symmetry, swapping (m, n) for $(-n, -m)$, cuts the casework in half.

Solution

The first result

Suppose $x, y > 0$ and $x^3 + x^2 = y^3 - y^2$, and suppose for contradiction that $x \geq y$. Since cubing and squaring are increasing on the positive reals,

$$x^3 + x^2 \geq y^3 + y^2 > y^3 - y^2,$$

where the strict step uses $y^2 > 0$. This contradicts the equation, so $x < y$.

The second result

Now let $0 < x \leq y - 1$, so $y \geq x + 1 > 1$. Write $f(t) = t^3 - t^2$. Then $f'(t) = 3t^2 - 2t = t(3t-2) > 0$ for $t > \frac{2}{3}$, so f is increasing on $[1, \infty)$. Since $y \geq x+1$ and both lie in that interval,

$$\begin{aligned} y^3 - y^2 = f(y) &\geq f(x+1) = (x+1)^2((x+1)-1) \\ &= x(x+1)^2 = x^3 + 2x^2 + x \\ &= (x^3 + x^2) + (x^2 + x) \\ &> x^3 + x^2, \end{aligned}$$

because $x^2 + x > 0$ for $x > 0$. So $x^3 + x^2 < y^3 - y^2$, as required.

Together, the two results say that for positive x, any positive y with $x^3 + x^2 = y^3 - y^2$ satisfies

$$x < y < x + 1.$$

No positive integer pair

Suppose positive integers a and b have $a^3 - b^3 = a^2 + b^2$. The right-hand side is positive, so $a^3 > b^3$ and $a > b$. Rearranging,

$$b^3 + b^2 = a^3 - a^2,$$

which is the equation above with $x = b$, $y = a$, both positive. So $b < a < b + 1$.

Ask yourself Where does integrality actually get used?

Only here: there is no integer strictly between b and $b + 1$. Put another way, $a > b$ with a, b integers forces $a \geq b + 1$, that is $0 < b \leq a - 1$, and then the second result gives $b^3 + b^2 < a^3 - a^2$, contradicting the equation. So no such pair of positive integers exists. (For real numbers there are plenty of solutions, one $y \in (x, x + 1)$ for each $x > 0$; the integers simply miss them.)

All integer pairs

We now want all integers m, n with

$$m^3 - n^3 = m^2 + n^2. \qquad (*)$$

Two observations save work.

- If (m, n) is a solution, so is $(-n, -m)$: indeed $(-n)^3 - (-m)^3 = m^3 - n^3$, and the sum of squares is unchanged.
- The right-hand side of $(*)$ is non-negative, so $m^3 \geq n^3$, that is $m \geq n$, with equality only if $m^2 + n^2 = 0$, i.e. $m = n = 0$.

One of them is zero. If $n = 0$, then $m^3 = m^2$, so $m = 0$ or $m = 1$. If $m = 0$, then $-n^3 = n^2$, so $n = 0$ or $n = -1$. This gives $(0, 0)$, $(1, 0)$ and $(0, -1)$ (the last is the image of $(1, 0)$ under the symmetry).

Both positive. Impossible, by the previous part.

Both negative. Then $-n$ and $-m$ are both positive, and $(-n, -m)$ would be a solution with both entries positive, which is impossible.

Opposite signs. Since $m > n$, we have $m > 0 > n$. Write $n = -p$ with $p \geq 1$. Then $(*)$ becomes

$$m^3 + p^3 = m^2 + p^2, \qquad \text{i.e.} \qquad m^2(m - 1) + p^2(p - 1) = 0.$$

For positive integers both terms are ≥ 0, so both are zero, forcing $m = p = 1$. This gives $(1, -1)$, and indeed $1 - (-1) = 2 = 1 + 1$.

So the integer solutions (m, n) of $m^3 - n^3 = m^2 + n^2$ are exactly

$$(m, n) = (0, 0), \quad (1, 0), \quad (0, -1), \quad (1, -1).$$

If "pair" is meant to exclude the trivial pair, drop $(0, 0)$.

Checks

Each pair can be checked by hand: for $(1, 0)$, $1 - 0 = 1 = 1 + 0$; for $(0, -1)$, $0 - (-1) = 1 = 0 + 1$; for $(1, -1)$, $2 = 2$. A computer search over all $|m|, |n| \leq 300$ finds exactly these four solutions. As a check on the first two results, numerically solving $y^3 - y^2 = x^3 + x^2$ for random $x > 0$ always gave $x < y < x + 1$; for example $x = 1$ gives $y \approx 1.696$.

Common slips

- In the final part, forgetting zero and negative integers, which is where all the solutions live.
- Assuming which of the two integers is "first". The difference of cubes is positive here, so the larger number must come first; say so.
- Proving the second result by just "expanding" without noting that $t^3 - t^2$ is increasing for $t \geq 1$; the step $f(y) \geq f(x + 1)$ needs it.

Beyond STEP

The argument "the real solution is trapped strictly between two consecutive integers, so there is no integer solution" is a standard tool in the study of Diophantine equations, and often the first thing to try on a polynomial equation in integers. Bounding a solution in this way is also how one proves many such equations have only finitely many solutions, which can then be found by a finite search. Davenport's *The Higher Arithmetic* is a lovely introduction to this kind of number theory.

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